

The University of the South Pacific

School of Information Technology, Engineering, Mathematics and Physics

IS328: Data Mining

Assignment I — Semester II, 2026

Total Marks: 20%

Due Date: 11:55 PM, Sunday 6th September 2026

This assignment covers both theoretical and practical aspects of the course. The marking rubric is aligned with Data & Information Management, consistent with the course outline and the BSE program map.

This assessment covers the following course learning outcomes:

- CLO 2: Perform pre-processing tasks to refine data sets
- CLO 3: Apply various data mining methods for interpreting results

Overview

The goals of this assignment are:

- To make you familiar with the WEKA tool and understand data preprocessing methods, attribute selection methods, classification, clustering algorithms (including an ensemble method such as Random Forest), and evaluation of models using the confusion matrix.
- Working in teams of 4 members, you are required to attempt all questions in this assignment. Your findings should be consolidated into a single report (including: applied algorithm, working screenshots and results, and analysis/comparison of outcomes).
- Prepare a clear, consolidated report that presents your findings.
- This assignment carries a total of 30 raw marks, which are scaled to 20% of your final grade. Marks will be awarded based on the completeness of findings, quality of analysis, and — importantly this year — the reasoning and justification behind the decisions you make, not only the steps you perform.

Grading

- Late submissions without prior notification and approval from the course coordinator will result in a 10% deduction per day.
- Plagiarism or cheating in any form is strictly prohibited. If detected, the entire Assignment 1 submission will be nullified.

Plagiarism

For all Assignment/Project works it is essential that you avoid plagiarism. Not only do you expose yourself to possibly serious disciplinary consequences, but you will also cheat yourself of a proper understanding of the concepts emphasized in the assignment.

- It is not plagiarism to discuss the assignment with your friends and consider solutions to the problems together. However, it is plagiarism if you copy all or part of each other's solutions.

AI Usage Permissions

Artificial Intelligence (AI) tools are restricted in this assessment. Students may use generative AI tools (e.g., ChatGPT, Copilot, or similar) only for non-technical tasks such as improving writing clarity, formatting the report, citation formatting, and basic background reading on a concept (e.g., what a decision tree is). Any use of AI must be clearly acknowledged.

Prohibited uses:

- Using AI for any technical or analytical task — including selecting or removing attributes, choosing parameters (e.g., the number of clusters k), running or interpreting WEKA output, comparing models, or writing the analysis, justifications, and recommendations. These are the tasks this assignment is designed to assess and must be your own work.
- Copying and submitting AI-generated content directly.

Acknowledgement:

- Students must document where AI was used (e.g., "AI was used to improve phrasing and format citations, which were then reviewed and verified by the team. No technical analysis or content was copied directly from AI outputs").

Rationale:

- This assessment develops your ability to independently preprocess, model, and interpret data mining results using WEKA. AI may support basic writing tasks but must not replace your own thinking or technical work. Undisclosed or prohibited AI use will be treated as academic misconduct under the plagiarism policy above.

Question 1: Data Pre-processing [5 Marks]

Dataset: You will use the Bank Marketing dataset (bank_dirty.csv). This dataset contains client demographic and campaign-contact attributes used to predict whether a client will subscribe to a term deposit (y = yes/no). Attribute descriptions are provided in the accompanying bank-names.txt file.

Use the bank data (bank_dirty.csv, available on Moodle) to perform a series of preprocessing operations using filters in WEKA.

A. Attribute Selection and Justification: [2 Marks]

Task:

- o Load the bank_dirty.csv file into WEKA and convert it into ARFF format.
- o **Decide and defend:** Examine all attributes using WEKA's visualizations and attribute statistics. YOU decide which attributes are irrelevant or redundant and should be removed — for example, consider the relationship between pdays, previous, and poutcome. There is no single correct answer; marks are awarded for the reasoning, not a specific list.
- o Remove the attributes you selected and save the dataset as bank_filtered.arff.

Expected Outputs:

- o A list of removed attributes, each with a clear justification for WHY it was removed (or why a candidate attribute was kept).
- o A screenshot of the WEKA interface showing the final list of attributes.

B. Discretization and Normalization: [2 Marks]

Task:

- o Discretize the attribute age into 3 bins (Young, Middle-aged, Senior).
- o Normalize the attribute balance to have values between 0 and 1.
- o Save the dataset as bank_preprocessed.arff.

Expected Outputs:

- o Screenshots showing the discretization and normalization steps.
- o Bin ranges and number of instances in each bin (for age).

C. Handling Missing Values: [1 Mark]

Task:

- o The bank_dirty.csv file contains missing values marked with "?" across several attributes (e.g. age, job, education, balance, contact, poutcome).
- o Apply the "ReplaceMissingValues" filter to handle these missing values.
- o Save the dataset as bank_missing.arff.

Expected Outputs:

- o A summary of how the missing values were handled.
- o A screenshot showing the missing values before and after replacement.

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Data and Information Management	I. Do not identify accurately any of the data quality problems	I. Identify accurately some of the data quality problems	I. Identify accurately most of the data quality problems	5	
	II. Do not perform all required tasks correctly and consistently	II. Perform most of the required tasks correctly and consistently	II. Perform all the required tasks correctly and consistently		
	III. Provide inaccurate and/or incomplete reports	III. Provide relatively accurate and complete reports	III. Provide accurate and complete reports		
Sub Total & comments					

Question 2: Classification Using J48, PART, Logistic Regression, and Random Forest [15 Marks]

A. J48 Decision Tree: [3 Marks]

Task:

- o Load the bank_preprocessed.arff dataset.
- o Apply the J48 algorithm to classify the dataset using 10-fold cross-validation, with y (yes/no) as the target (class).
- o Save the model and analyze the resulting decision tree.

Expected Outputs:

- o Accuracy of the model and structure of the decision tree.
- o Screenshot of the J48 tree in WEKA.
- o A brief comment on interpretability (Is the tree easy to read?).

B. PART Rule Analysis: [3 Marks]

Task:

- o Apply the PART algorithm with 10-fold cross-validation, ensuring y is the target attribute.
- o Save the model and compare the results with J48.

Expected Outputs:

- o Rules generated by PART.
- o Comparison with J48 in terms of accuracy, simplicity, and interpretability.

C. Logistic Regression: [3 Marks]

Task:

- o Apply Logistic Regression to the dataset with 10-fold cross-validation.
- o Compare Logistic Regression's accuracy and performance against J48 and PART.

Expected Outputs:

- o Logistic Regression accuracy and coefficients summary.
- o Discussion on whether Logistic Regression handles categorical and numerical attributes effectively compared to tree-based models.

D. Analysis of Misclassification: [3 Marks]

Task:

- o Use WEKA's "Visualize errors" option in the Classify tab to identify instances where J48, PART, and Logistic Regression misclassified the subscription outcome.
- o Look for common patterns in misclassified instances (e.g., age, job, balance, contact duration).

- o Suggest possible preprocessing improvements (e.g., feature engineering, binning continuous attributes).

Expected Outputs:

- o Screenshots of misclassification results.
- o A short discussion summarizing the patterns and improvements.

E. Random Forest (Ensemble Method): [3 Marks]

Task:

- o Apply the Random Forest algorithm to the dataset with 10-fold cross-validation.
- o Compare the performance of Random Forest with J48, PART, and Logistic Regression using accuracy, precision, recall, and F1-score.

Expected Outputs:

- o Table of performance metrics (Accuracy, Precision, Recall, F1).
- o Brief discussion on whether Random Forest improved over individual classifiers.

Question 3: Clustering and Analysis [5 Marks]

A. Applying k-Means Clustering: [3 Marks]

Task:

- o Use the bank_preprocessed.arff dataset.
- o **Decide and defend:** Run k-Means for at least two different values of k (for example k=2 and k=3). YOU decide which value of k is most appropriate for this dataset and justify your choice — you may refer to the within-cluster sum of squared errors, cluster sizes, or interpretability. Note that k=2 is an obvious starting point since y has two categories, but you must argue whether it is actually the best choice.
- o Using your chosen k, save the clustered dataset as bank_clustered.arff.

Expected Outputs:

- o Screenshots of the k-Means results for the values of k you tried.
- o A clear justification of your chosen k (referring to SSE, cluster sizes, or interpretability).
- o A short discussion on what the clusters represent (e.g., which attributes such as age, job, balance, duration appear to drive cluster formation).

B. Comparing Clusters with Classification Results: [2 Marks]

Task:

- o Compare the cluster assignments from k-Means with the actual subscription labels (y).
- o Discuss alignment or discrepancies between the unsupervised clusters and the supervised classification results from J48, PART, Logistic Regression, and Random Forest.

Expected Outputs:

- o A brief comparison highlighting whether the clusters overlap meaningfully with the two subscription groups.
- o A discussion on why clustering may or may not align with classification (e.g., clusters may capture job/balance patterns but not subscription directly).

Question 4: Advanced Evaluation Metrics [5 Marks]

A. Evaluation Using Confusion Matrix: [3 Marks]

Task:

- o Evaluate the models (J48, PART, Logistic Regression, Random Forest) using their confusion matrices in WEKA.
- o Compare the results across models in terms of Accuracy, Precision, Recall, and F1-score.
- o Save and include the evaluation results.
- o **Critical thinking:** This dataset is imbalanced — the great majority of clients did NOT subscribe (y = no). Using your confusion matrices, argue whether accuracy is a fair metric for this problem. Would a model that predicts "no" for every client score a high accuracy? Would it be useful to the bank? Explain which metric(s) you would trust instead and why.

Expected Outputs:

- o Confusion matrices for all models.
- o A summary table comparing Accuracy, Precision, Recall, and F1-score for each model.
- o A reasoned argument on whether accuracy is appropriate here given the class imbalance, and which metric best identifies clients who WILL subscribe.
- o A brief note on which model is the most reliable for predicting the subscription outcome (yes/no), justified by the metrics you chose to trust.

B. Practical Recommendations: [2 Marks]**Task:**

- o **Evidence-based:** Based on your analyses from the classification and clustering tasks, provide practical recommendations for how a bank's marketing team could improve term-deposit subscription rates and target campaigns more effectively. Each recommendation **MUST** be tied to specific evidence from your results (e.g., a decision-tree split, a PART rule, a cluster profile, or an attribute's influence) — general advice not grounded in your findings will not earn marks.
- o Focus on the most influential attributes and suggest actionable, specific steps the marketing team could implement.

Expected Outputs:

- o A short report with recommendations supported by data analysis.

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Data and Information Management	IV Do not use correct algorithm(s) for the problem in hand	IV Identify and use correct algorithm for the problem in hand	IV Identify and use correct algorithm for the problem in hand	25	
	II Do not perform all required tasks correctly and consistently	II Perform most of the required tasks correctly and consistently	II Perform all required tasks correctly and consistently		
	III Provide inaccurate and/or incomplete reports	III Provide relatively accurate and complete reports	III Provide accurate and complete reports		
Sub Total & comments					

Submission Instructions

1. Completely fill the Mark Allocation Sheet and submit with your assignment. Failing to do so may result in a deduction of 50% of marks.
2. This assignment can be submitted in groups of 4 members. Assign a group leader and submit the assignment through the group leader's Moodle account. Submit just one zip/rar file of your project. The submission filename should read A1_Sxxx_Syyy_Szzz_Swww.pdf or A1_Sxxx_Syyy_Szzz_Swww.docx where Sxxx_Syyy_Szzz_Swww are the student IDs of the group members. Incorrect submission will result in a high penalty.
3. 25 Marks are allocated for applying appropriate DM techniques and deriving the correct results for the drafting and consolidation of the report.

Mark Allocation Sheet

After having discussed as a group, we recommend the following mark allocation to each group member based on contribution (or lack of it) throughout the assignment.

Group Name		
Project Manager		
Member ID and Name	Percentage contribution of allocated task	Signature

Assessments mapping with CBOK

Core Body of Knowledge		IS328	Test1	Assign 1	MST	Presentation	Assign 2
	Complex Computing						
ICT Professional Knowledge	Ethics	M					
	Professional expectations	M	✓		✓		
	Teamwork concepts/issues	M				✓	✓
	Communication	M				✓	✓
	Societal Issues/Legal issues/Privacy	M					
	Understanding the ICT profession						
ICT Problem Solving:	Abstraction						
	Design						
Technology Resources	Hardware and Software Fundamentals						
	Data and Information Management	M		✓			✓
	Networking						
Technology Building	Human Factors						
	Programming						
	Systems Development / Acquisition						
ICT Management	IT Governance and organisational issues						
	IT Project management						
	Service management						
	Security management						